Automating Analysis of Fast-Camera Footage of Plasma Instabilities in PFRC-2

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Abstract

Rotational mode instabilities within the plasma of the PFRC-2 have the potential to destabilize the FRCs within the central chamber of the device and disrupt plasma confinement. These instabilities in plasma can be captured as fast camera footage and slowed down significantly to be studied in detail. When investigated alongside varied machine parameters, it would be possible to determine how physical changes made to the system impact its overall stability. This paper aims to introduce and explore a unique approach for examining plasma motion by automating, and thereby streamlining, the analysis process. In this approach, trends in spatial and temporal intensity variation were analyzed to discern and classify the types of motion occurring within the plasma. The algorithm detailed in this report was able to successfully categorize plasma motion with reasonable (>90%) accuracy.

1 Introduction

The Princeton Field-Reversed Configuration (PFRC-2) is a prototype of a small-scale, low-neutron-emission nuclear fusion reactor concept. It utilizes sustained field-reversed configurations (FRCs) that consist of closed field lines trapping energy and confining plasma to enable nuclear fusion. As a result, the plasma forms a self-stable torus elongated along the central axis of the device. The prototype uses a single 1.8 MHz rotating field system known as an odd-parity rotating magnetic field (RMF) to drive plasma current, enable heating and maintain stability for pulses lasting up to ∼300 ms.

Theoretical calculations involving magnetohydrodynamic models have predicted that tilt modes, plasma instabilities that occur due to tilting of internal magnetic fields causing a loss of confinement, arise on a time scale of ∼1 µs. However, plasma discharges have been shown to survive for up to 300 ms, several magnitudes greater than the time scale of the tilt instabilities that would have caused the FRCs to collapse. Despite this, the plasma does still display rotational mode instabilities. When the plasma is viewed along the central axis of the device, these rotational modes can be seen as bright (high intensity) regions rotating around the center of the cylindrical central cell chamber. These rotational instabilities were observed to shift form with varying plasma parameters. The behaviour of these plasma instabilities is crucial to understanding the stability system. One such example would be the presence of stationary ("locked") modes within the plasma. In a tokamak reactor, locked modes could disrupt plasma rotation and degrade plasma confinement, while in a PFRC reactor, locked modes would indicate increased stability within the plasma. Evidently, it is essential to study how rotational mode instabilities vary within the plasma to better control and stabilize the plasma and the system as a whole.

The movement of the plasma and its instabilities within the central cell chamber of the PFRC-2 was observed for each set of machine parameters through a Vision Research Phantom v7.3 High Speed camera ("fast-camera"). The motion was captured into a digital file, slowed down and then analyzed by eye. While this manual technique is more likely to yield accurate results, it can be cumbersome and prone to human error when fast-camera video files need to be analyzed for larger quantities of experimental runs. This project aims to address this concern and streamline the process of analyzing fast-camera footage by constructing an algorithm using MATLAB (v23.2) code with a preliminary goal of automating specific components of central cell plasma motion categorization.

2 Method

In the captured video files, bright projections could be observed that resemble spokes extending between the walls of the central cell chamber and central axis of the machine. These plasma spokes either remain stationary (i.e. 'locked') or rotate around a central point. The algorithm serves to identify two properties of plasma as viewed through the camera: the direction of spoke rotation (clockwise or counter-clockwise), if any, and the presence of locked spokes. Each media file is composed of frames of resolution 64 × 64 pixels numbering between ∼500 and 5000 corresponding to the RMF pulse length. The interval between frames is 7.14 µs and exposure time is 1 µs. Computation of pixel brightness across frames will yield temporal intensity variation while analysis of pixel brightness at different positions on a given frame will yield spatial intensity variation. As the spokes appeared to extend toward the center of the vessel and rotate around the central axis, the intensity of pixels at a fixed radius from the central point (i.e. forming a circle) were sampled and compared to compute angular intensity variations. Lastly, to construct the algorithm and conduct preliminary testing, data from 59 experimental runs conducted on the PFRC-2 across three separate days (07/17/2025, 08/05/2025, 08/15/2025) was utilized.

Expanding upon the computation of intensity variation, angular intensity variations were calculated using 36 predetermined pixels (each 10° apart, ± 1 pixel) positioned at a radius of 28 pixels from the center of the circular cross-section of the chamber. The position of the center relative to the edges of the frames varied slightly (up to 4 pixels in one direction) across different days (experimental sessions). Therefore, a function was constructed to simplify and automate the process of identifying the central coordinate. This function considers the earliest video of an experimental session and determines the mean intensity value averaged across all frames in the video for each pixel situated in a 8 × 8 central square (defined by edge coordinates (28, 28), (28, 35), (35, 28), (35, 35)) that spans 1.56% of the frame's area. The pixel with the highest averaged intensity is established as the center of the chamber for all fast-camera videos captured during the same experimental session. The intensity values for the 36 circumferential points computed with reference to this central point are sampled for every frame and stored in a 2D matrix with dimensions being number of points (36) and number of frames (dependent on video). Angular and temporal trends in intensity are discerned through analysis of this data matrix, which are then used to categorize plasma motion.

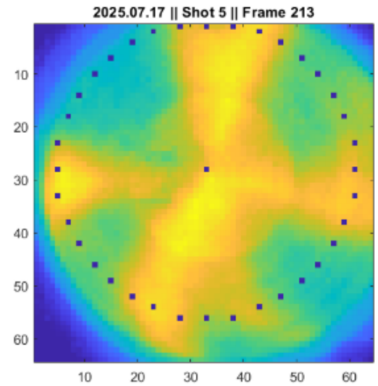

Figure 1: 36 preselected circumferential points in 64 × 64 frame

2.1 Presence of Locked Modes

Angular variation of pixel intensity around the central axis of the chamber is utilized to evaluate the presence of locked spokes in the fast-camera footage. It was observed that the average brightness varied temporally, so the 36 intensity values associated with each frame were normalized such that mean intensity across these 36 points was 0.00 and standard deviation was 1.00. It was also noted that, in certain experimental runs, locked modes formed, shifted slightly and collapsed over the course of the video. Therefore, the video was divided into 1 ms sections (140 frames) and the presence of locked modes was computed separately for each millisecond. The intensity value at each circumferential point during a given millisecond interval was calculated by averaging the intensity values of the specific point across all 140 frames within the 1 ms period. When this is performed for the complete video for all 36 points, the resulting values form a temporally compressed version of the previous raw angular-temporal matrix (array).

When the 36 averaged intensity values in a given millisecond interval are plotted against angular position, trends of trigonometric nature are observed. This is consistent with predictions because stationary spokes would appear as brighter regions with higher intensity while regions on either side of a spoke would appear much darker with lower intensity. When traversing around the center of the chamber, this would form a pattern with alternating peaks and troughs of brightness, resembling a sine or cosine wave. To more accurately determine such patterns, a Fast Fourier Transform (FFT) algorithm was applied to the values. The FFT output would be such that the discrete frequency values (in per frame, frame^{-1} = 0.14 MHz) would correspond to the number of spokes (i.e. number of stationary bright regions around the center). The magnitude value associated with each frequency indicates the amplitude of the wave component at that frequency. For a given frequency (number of spokes), the amplitude indicates how strongly this frequency appears in the intensity data. Therefore, a higher amplitude corresponds to a greater presence of locked modes, indicating a greater likelihood of locked modes being observed.

When the FFTs were plotted and analyzed, a peak amplitude appeared at a frequency of 4 (corresponding to 4 spokes) for almost all videos in which locked modes were observed. A peak was also observed at a frequency of 1 in some videos that was disregarded, likely caused by the presence of a spoke that is significantly brighter than the remaining spokes. Therefore, to establish a consistent procedure for determining the locked-ness of plasma spokes in a fast-camera video, it was decided that the magnitude of the FFT at frequency 4, which represents the intensity of 4 spokes, would be used to deduce the likelihood of observing locked modes. Of the 59 experimental runs this procedure was tested on, 57 were categorized as "Locked" or "Not Locked" correctly (matching observations made by eye) when a threshold value of 5.0 (arbitrary intensity units) was used for the presence parameter. This corresponds to a 96.6% initial success rate and indicates that our algorithm, when used to detect locked modes, is able to provide results with a high degree of accuracy.

2.2 Direction of Spoke Rotation

The method used to compute the direction of rotation of spokes around the central axis of the chamber closely mirrors the method used to compute locked mode presence. The key differences between the procedures are the use of temporal intensity variations in place of spatial intensity, and the manner in which the final fast Fourier transform output is utilized to classify the plasma motion. Another difference is that the direction is computed for the entire fast camera video, instead of 1 ms sections of it. The latter facility could be added to the algorithm without much alteration, but a decision was made on the final algorithm to analyze rotation over the entire video duration. This is due to the fact that many videos displayed unstable "wobbling" motion of spokes beyond just the overall spoke rotation about the center. These "wobbles" were short-lived relative to the broader rotation of plasma spokes, but over smaller time durations such as the 1-ms-sections, their effects would become more significant and could potentially disrupt the algorithm's overall analysis of the direction of spoke motion in a video. Considering plasma spoke motion over the maximum length of time (duration of full video) would minimize the effects of smaller "wobbles".

It was observed that the intensity of spokes fluctuated as they moved around the central axis, and the overall brightness of frames varied across the full length of the video. To reduce the effects of these fluctuations on the fast Fourier transform output, the intensity values from all frames at each point around the center were temporally normalized. Essentially, each of the 36 preselected points was considered individually and the intensity values at that specific point for all frames in the video were passed through a normalization process that adjusted the values such that their mean would be 0.00 and their standard deviation would be 1.00. This temporally normalized row of intensity values at a given preselected point was passed through an FFT algorithm and the phase values were calculated from the subsequent complex output. Hence, the final output values describe the phase position of a wave component of a specific frequency. When compared against the phase values of neighboring points (points around the center that are adjacent to the point being considered), we can determine the direction of motion of the wave components between the two points. If two points are compared and the point in the clockwise direction has a greater phase value at a given frequency than the point in the counter-clockwise direction, the wave would have encountered the clockwise point first and then reached the the counter-clockwise point, meaning it traveled counter-clockwise between the two points. If the point in the clockwise direction has a lower phase value at a given frequency than the point in the counter-clockwise direction, the wave component has traveled clockwise between the two points. Hence, we can compare the phase values of points against that of adjacent points along the edge of the chamber for each discrete frequency wave component until the Nyquist limit. The direction predictions from all pairs of adjacent points across all frequencies were considered and the direction ("clockwise" or "counter-clockwise") predicted most frequently across all frames of the video was chosen as the overall predicted direction for the video. When these final predictions were computed for the 59 videos, it was observed that, in general, "counter-clockwise" was predicted slightly more frequently than "counter-clockwise". In videos classified as clockwise by eye, the percentage of clockwise predictions varied around 60% and was, in a few, as low as 51%, whereas the lowest percentage of counter-clockwise predictions for videos classified by eye as such was 54%. It was also noted that increasing the exposure level of the fast camera typically improved prediction accuracy for clockwise runs. During the initial development of the algorithm, it was suggested that only predictions from frequencies at which the magnitude of the corresponding wave component peaked (i.e. peaks in graph of FFT magnitude against frequency) should be considered when determining the overall direction prediction. However, of the 59 runs that were used in the construction and preliminary testing of the algorithm, only 5 runs displayed any substantial peaks in magnitude when plotted against frequency. Therefore, the final revision of the algorithm was modified to consider predictions at all frequencies. Similar to the earlier method for locked mode detection, specific thresholds were established to distinguish between clockwise, non-directional/stationary and counter-clockwise plasma motion as accurately as possible. If the portion of clockwise predictions exceeded 50.5%, the plasma motion would be classified as "clockwise" and if the portion of counterclockwise predictions exceeded 54% (i.e. $< 46\%$ clockwise), it would be classified as "counter-clockwise". If the predictions failed to satisfy both conditions (i.e. is between 46% and 50.5% clockwise), the run would be classified as just "non-directional". This specific set of thresholds was chosen as it minimized error to 96.6%

(57 of 59) for the 59 runs used as preliminary test data. Therefore, using the aforementioned techniques and this set of thresholds, the algorithm is able to successfully differentiate between clockwise, counter-clockwise and stationary plasma motion with a high degree of accuracy.

3 Discussion

The two qualities classified by the algorithm are not necessarily independent of each other but are not fully interlinked either. The direction of spoke motion was categorized as stationary if there was no substantial motion detected in either direction or if no distinguishable spokes were present. However, the detection of locked modes may have been disrupted by the complete absence of spokes in a video. Hence, it is important to consider both aspects of plasma motion classification when analyzing the results generated by the algorithm to identify such anomalies.

3.1 Analysis of Results

The data collected during the 07/17/2025 experimental session consisted of 33 runs on the PFRC-2 during which fast camera footage of plasma motion within the central cell was captured. Of all the considered data sets, this set had the most variation in terms of plasma motion. The values for locked mode presence (arbitrary units) and plasma rotation direction (fraction of clockwise predictions) were plotted against each other as shown in figure x.

The locked mode presence value ("locked-ness") and direction value can be used to determine the overall motion of the plasma within the central cell. Figure 2 shows how these two values vary in relation to each other. The points appear to approximate a bell curve with a mean of 0.481 that passes very close to the intersection points of the two sets of thresholds. This is consistent with what we would expect because the midpoint of the thresholds for differentiating direction is 0.48, so this should be where the presence of stationary locked modes is highest. As the direction value increases or decreases away from 0.48, the plasma begins to rotate in one direction so the presence of locked modes reduces, as seen on the graph. From this, we can deduce that the values output from the algorithm are reasonable and valid as they follow this expected general trend.

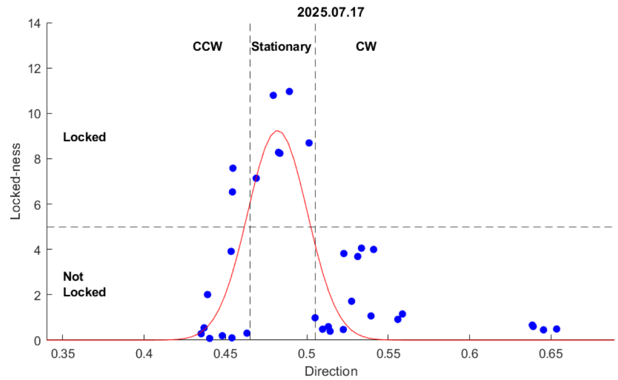

Figure 2: Numerical values for direction and presence of locked modes in 07/17/2025 data

3.2 Analysis of Trends

Though determining motion of plasma by eye will always be more accurate, there are many reasons why having a numerical value to represent motion may be beneficial. One such benefit is that it can provide a means to compare and contrast different motions that, to a human eye, may not be easily distinguishable. This would allow for more accurate determination of how different aspects of motion change with other parameters. An example of this can be seen in the 08/05/2025 data (shown in figure 3) where the magnetic flux density over the central axis (B) was varied across experimental runs.

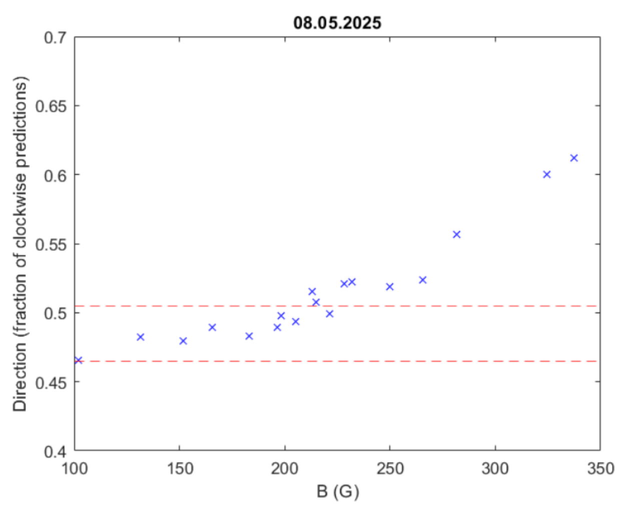

Figure 3: Change in ratio of clockwise predictions with B field values 08/05/2025 data

It can be seen that as the value of B increases from 100 to 350 G, the fraction of clockwise predictions, which indicates the direction, increases consistently. This would mean the motion of the plasma spokes with increasing B becomes more clockwise overall and the hidden counter-clockwise motions (such as "wobbling") reduce. The transition between distinguishable overall clockwise motion and overall stationary (locked) motion appears to occur between 200 to 220 G. This observation would have been much more challenging to make in the absence of a numerical quantity for direction.

Another crucial benefit of this algorithm is that it can compute large numbers of runs using the same standards and criteria for differentiation, which is essential in ensuring plasma instability motion classifications are consistent. This consistency would be difficult to maintain if there are multiple users working with the data or if one user is studying a large number of runs. Therefore, automating the process of spoke motion classification allows for the establishment of a fixed standard applicable to all runs. An instance of this was observed when the algorithm was applied to 46 runs from 08/06/2025, a dataset that was not used during the construction of the algorithm. The algorithm was able to categorize all 46 videos in under one minute.

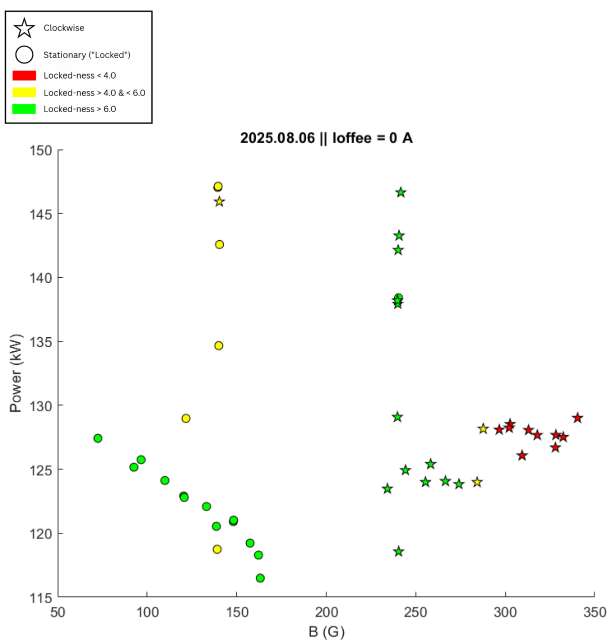

Figure 4: Categorized 08/06/2025 data

The figure (Fig. 4) shows the categorization of the 46 videos based on the direction of overall plasma motion and the presence of locked modes. Once again, the transition between visibly clockwise and visibly stationary motion appears to take place at B field values of ∼200 G. It can also be observed that for lower B fields, increases in power reduced the presence of locked modes. The use of the algorithm allows for these data files to be classified swiftly and based on a consistent, unsubjective metric.

4 Conclusion

The algorithm detailed in this report aims to accurately classify the motion of plasma and rotational mode instabilities within the central cell chamber of the PFRC-2. Utilizing patterns in angular and temporal intensity variation, the algorithm was able to classify the motion of the plasma projections based on direction ("clockwise", "stationary", "counterclockwise") and presence of locked modes. The algorithm was able to achieve an accuracy of 96.6% across the 59 runs used for preliminary testing. While observations of the video made by eye can better identify anomalies and erroneous data points (such as the complete absence of spokes), the algorithm is able to assist in reducing human error/bias, analyzing large quantities of video files, applying a consistent metric for classification, and quantifying otherwise purely qualitative measurements. Future studies can work to progress this algorithm by incorporating other mathematical models and numerical analysis techniques to more accurately categorize the data. Overall, this algorithm presents a unique approach for analyzing the motion of plasma and its instabilities by working towards automating the process. The analysis streamlined by this algorithm would help inform our understanding of instabilities within the PFRC-2 and, eventually, our ability to control these same instabilities.

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